MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5103 Linux OS and Scripting Theory – 3

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Linux Filesystem

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

1. The storage devices available.

The available storage device on the Linux system is /dev/sda.

2. The partitions available on each device.

The available partitions on /dev/sda are /dev/sda1 and /dev/sda2.

3. The filesystem type used by each partition.

The filesystem types used are Ext4 for /dev/sda1 and swap for /dev/sda2.

4. The device/partition on which `/` is located.

The root directory / is located on the partition /dev/sda1.

5. The filesystem type used for `/`.

The filesystem type used for the root / is Ext4.

6. The total, used, and available storage space.

The root partition has a total size of 19G, while the used and available storage space can be checked using the df -h / command.

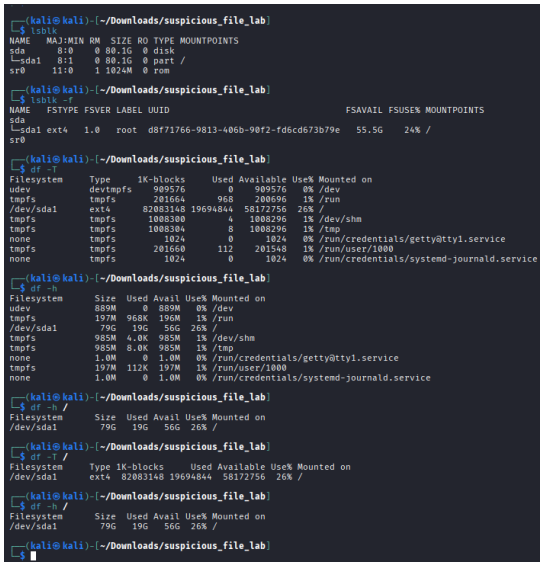
Suggested commands:

lsblk

lsblk -f

df -T

df -h



Device/Partition	Filesystem Type	Size	Mount Point
/dev/sda1	Ext4	19G	/
/dev/sda2	swap	2G	[swap]
tmpfs	tmpfs	~1.6G	/run
tmpfs	tmpfs	~7.7G	/dev/shm
tmpfs	tmpfs	~5.0M	/run/lock
tmpfs	tmpfs	~1.6G	/run/user/1000

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

Linux Storage Organization

/dev/sda

|— /dev/sda1 ext4 ~19G mounted at /

|— /dev/sda2 swap ~2G used as swap

The physical disk sda is divided into separate partitions such as sda1, sda2, The sda1 partition uses the ext4 filesystem and is mounted at /, while sda2 uses the swap filesystem for swap space.

Exercise 3 – Explore the Root Filesystem

Start from your current location.

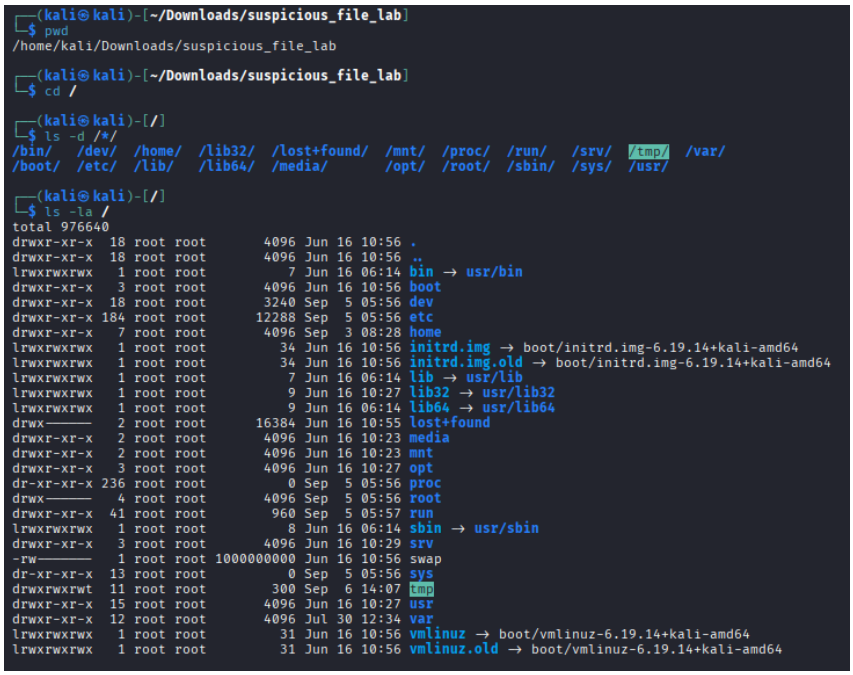
1. Determine your current working directory.

2. Navigate to the root directory.

3. Display the contents of the root directory.

4. Identify all directories directly available under `/`.

5. Determine whether hidden entries are present.



Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations:

For each location:

1. Examine its contents.

2. Identify at least three entries, where available.

3. Determine the likely purpose of the directory.

Directory	Example Contents	Purpose
/home	User folders, documents	Stores personal files of users.
/root	Root user's files	Home directory of the root user.
/etc	Configuration files	Contains system and application configuration files.
/var	Variable data, logs	Stores data that changes during system operation.
/var/log	System and application logs	Stores log files.
/tmp	Temporary files	Stores temporary files created by users and applications.
/usr	Programs, libraries, documentation	Contains most user applications and system utilities.
/boot	Kernel, bootloader files	Contains files required to boot the Linux system.
/dev	Device files	Provides access to hardware devices.
/proc	Process and system information	Provides information about running processes and the kernel.

Exercise 5 – Find Your Home Directory

Without assuming its location:

1. Determine your username.

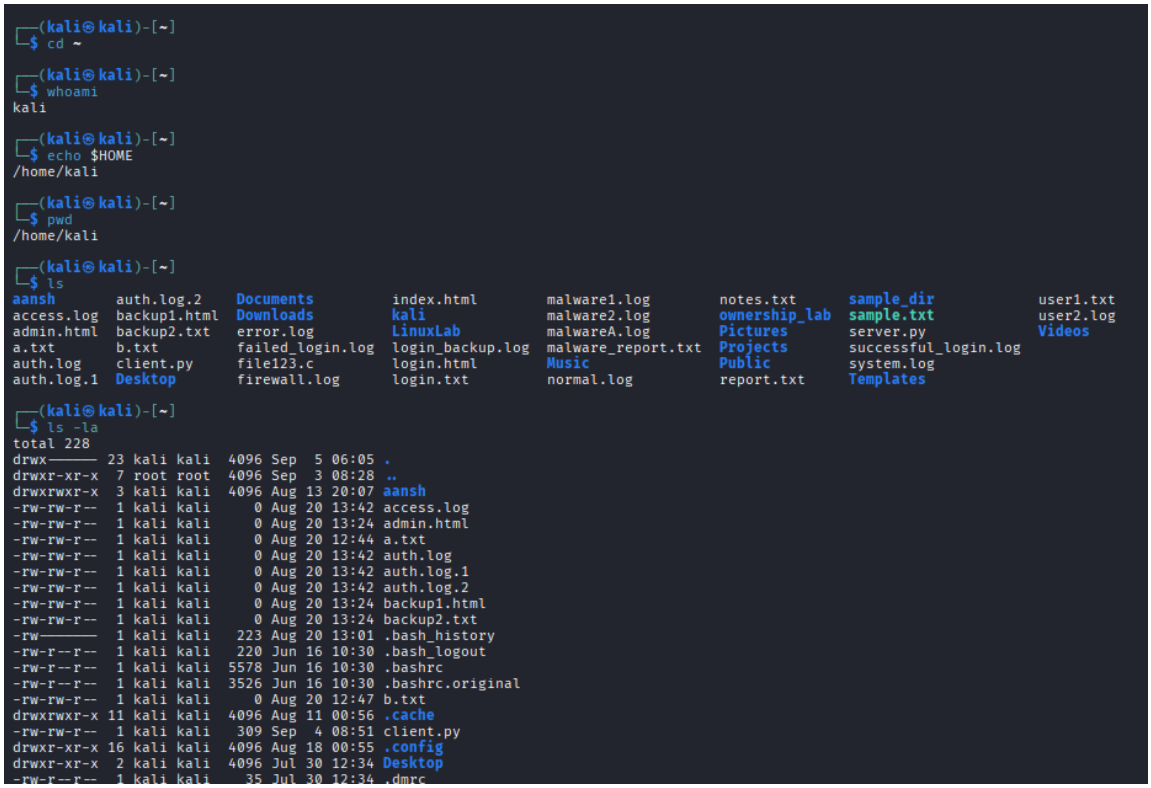
2. Determine your home directory.

3. Navigate to your home directory.

4. Display its absolute path.

5. List its normal contents.

6. List all contents, including hidden files.



Exercise 6 – Hidden Files Investigation

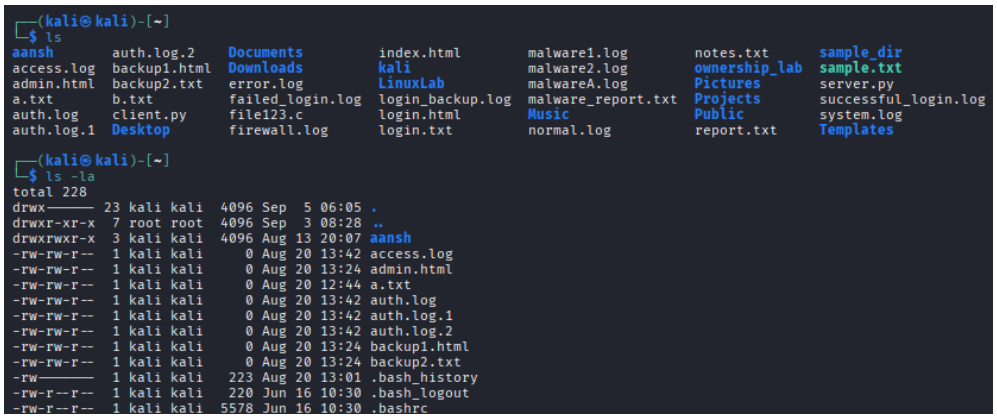
Examine your home directory.

Compare:

ls

ls -a

Identify at least 2 hidden files/directories.



Answer:

1. How does Linux identify a hidden file?

A hidden file in Linux is identified by a filename that begins with a dot (.).

2. Why does a normal `ls` not display these files?

A normal ls command does not display hidden files because it hides filenames beginning with a dot by default.

3. What do `.` and `..` mean?

. represents the current directory, while .. represents the parent directory.

4. Can a hidden file still be accessed if its name is known?

Yes, a hidden file can be accessed if its name is known by specifying its name or path.

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden files should not automatically be considered safe or malicious because they can be legitimate system/configuration files or potentially contain malicious content.

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: mkdir

Hint: you can create this hierarchy in a single command

Create the following files:

cyber_lab/evidence/network/traffic.txt

cyber_lab/evidence/system/processes.txt

cyber_lab/logs/security.log

cyber_lab/scripts/check.sh

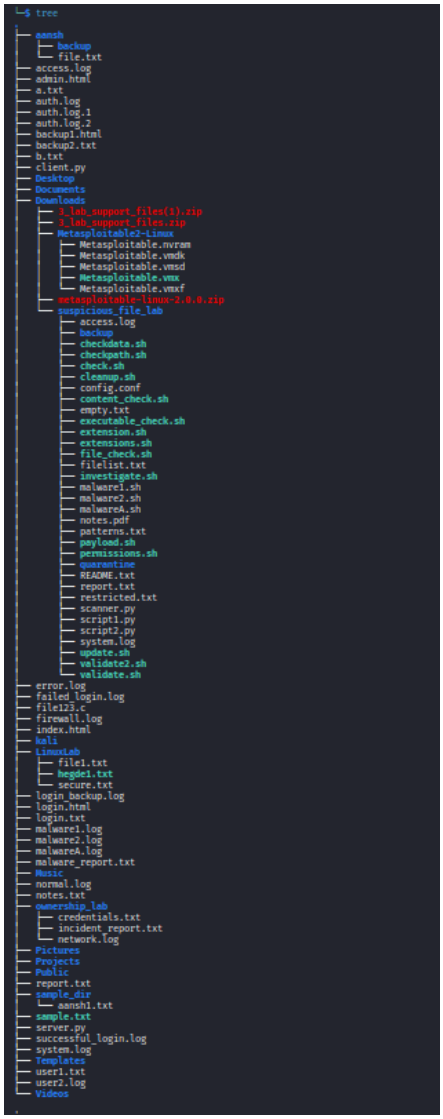
cyber_lab/reports/investigation.txt

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

Hint: tree

Explore options



Exercise 8 – Absolute and Relative Paths

Assume your current directory is:

~/cyber_lab

Perform the following:

1. Navigate to `evidence/network` using a relative path.

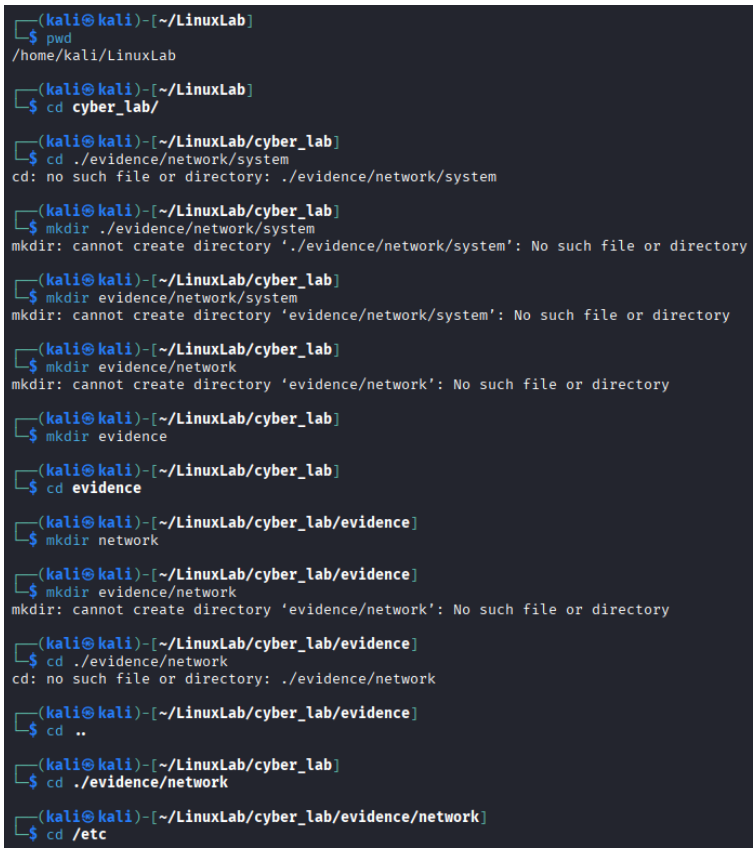
2. Navigate to `/etc` using an absolute path.

3. Return to your home directory using the shortest possible command.

4. Navigate to `cyber_lab/reports`.

5. Move one directory upward.

6. Display the current directory after every operation.



What is an absolute path?

An absolute path starts from the root directory / and gives the complete location of a file or directory.

What is a relative path?

A relative path specifies a location based on the current working directory.

What is the difference between `/` and `~`?

/ : root directory of the entire Linux filesystem.

~ : shortcut for the current user's home directory.

Exercise 9 – Identify File Types

Investigate the following:

/etc/passwd

/etc

/bin/bash

/dev/null

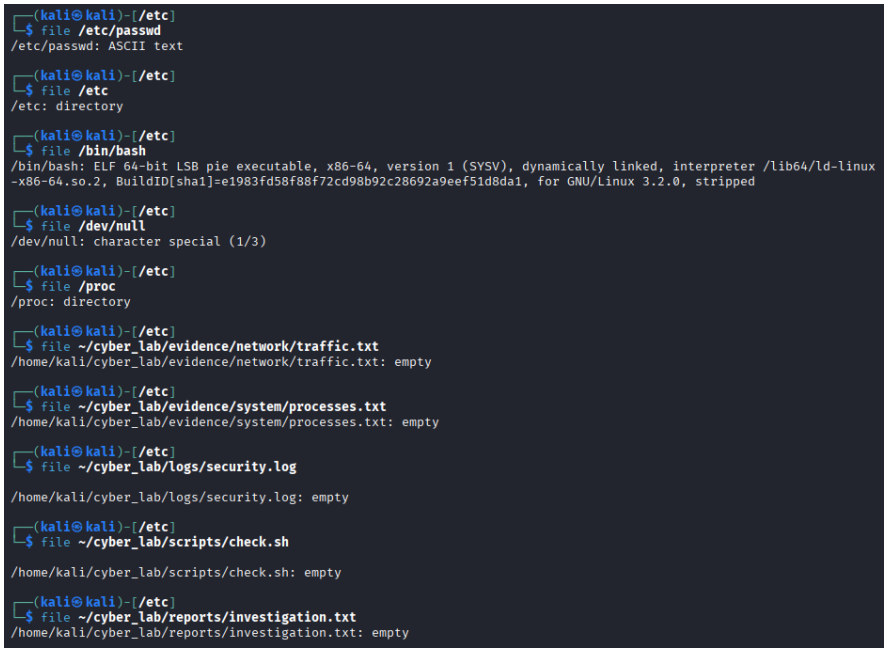
/proc

Also investigate the files created in Exercise 7.

file <filename> ls –

l<filename>

Classify each object.



Object	Type
/etc/passwd	Regular text file
/etc	Directory
/bin/bash	Executable file
/dev/null	Character special device file
/proc	Virtual/pseudo filesystem directory
traffic.txt	Regular file
processes.txt	Regular file
security.log	Regular file
check.sh	Regular file
investigation.txt	Regular file

Exercise 10 – Examine File Metadata

Use: ~/cyber_lab/reports/investigation.txt

Add a small amount of text to the file.

Examine it using:

ls -l

stat investigation.txt

Identify:

*** Filename**

*** File size**

*** File type**

*** Owner**

*** Group**

*** Inode number**

*** Access time**

*** Modification time**

*** Change time**

Answer:

```
┌──(kali㉿kali)-[/etc]
└─$ cd ~/cyber_lab/reports

┌──(kali㉿kali)-[~/cyber_lab/reports]
└─$ ls -l investigation.txt
-rw-rw-r-- 1 kali kali 21 Sep  6 15:13 investigation.txt

┌──(kali㉿kali)-[~/cyber_lab/reports]
└─$ stat investigation.txt
  File: investigation.txt
  Size: 21              Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 1179899     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/    kali)   Gid: ( 1000/    kali)
Access: 2026-09-06 14:57:36.852777525 -0400
Modify: 2026-09-06 15:13:28.480692737 -0400
Change: 2026-09-06 15:13:28.480692737 -0400
 Birth: 2026-09-06 14:57:36.852777525 -0400
```

1. What is metadata?

Metadata is information about a file, such as its size, owner, permissions, inode number, and timestamps.

2. Is metadata the same as file content?

No, metadata is not the same as file content; metadata describes the file, while content is the actual data stored inside it.

3. Which timestamp changes when file contents are modified?

The modification time (mtime) changes when the contents of a file are modified.

Exercise 11 – Explore `/etc`

Investigate:

/etc

```

**Locate:**

**passwd**

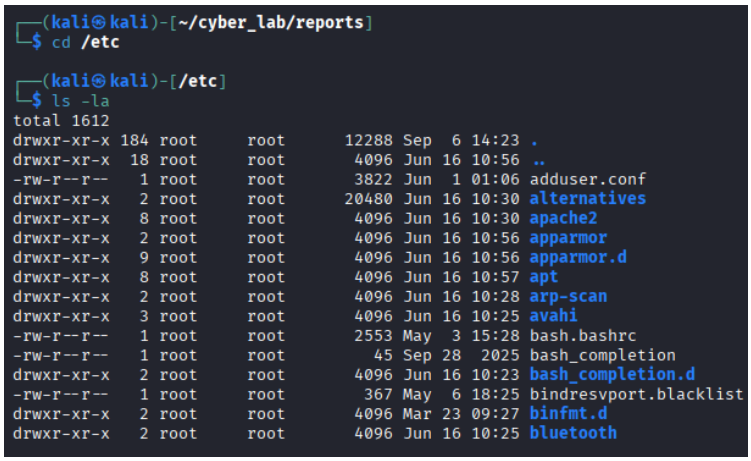
**group**

**hostname**

**hosts**
```

Note: Without modifying any system files, examine their contents where permitted.

Answer:



1. What kind of information is primarily stored under `/etc`?

The /etc directory primarily stores system-wide configuration files and settings.

2. What information does `/etc/passwd` contain?

The /etc/passwd file contains information about user accounts, such as usernames, user IDs (UIDs), group IDs (GIDs), home directories, and login shells.

3. What information does `/etc/hostname` contain?

The /etc/hostname file contains the hostname assigned to the Linux system.

4. Why is `/etc` security-sensitive?

The /etc directory is security-sensitive because it contains important system configuration, user-account, authentication, and network configuration information.

Exercise 12 – Explore Linux Log Storage

Investigate:

/var/log

```

**Identify:**

**Available log files**

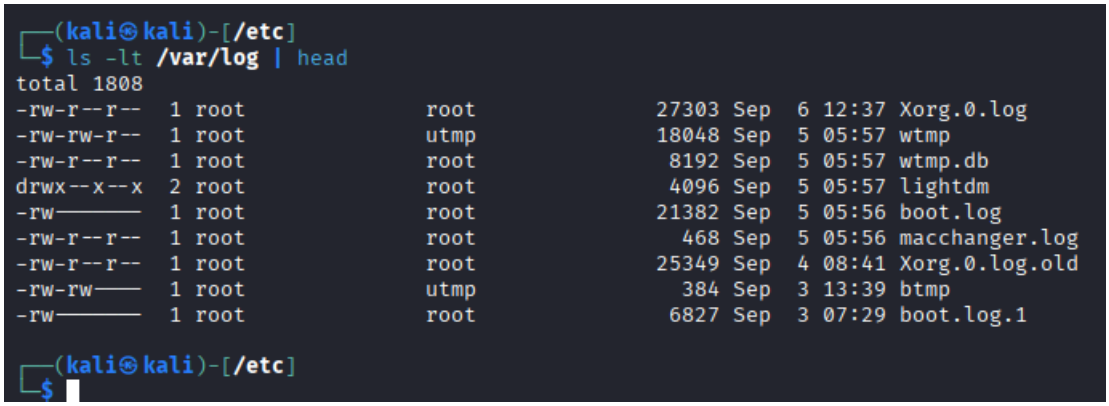
**Available log directories**

**Recently modified entries**

**Entries you can read**

**Entries you cannot read as a normal user**

**Answer:**
```



- **Available log files:** Xorg.0.log, wtmp, wtmp.db, lightdm, boot.log, macchanger.log, Xorg.0.log.old, and btmp.
- **Available log directories:** /var/log contains log files and directories such as lightdm.
- **Recently modified entries:** Xorg.0.log is the most recently modified entry shown.
- **Entries you can read:** Files with read (r) permission for your user or group can be read.
- **Entries you cannot read as a normal user:** Files owned by root without appropriate read permissions cannot normally be read.

1. **Why does Linux maintain logs?**
   Linux maintains logs to record system and user activities, errors, and events.
2. **Why are logs useful during incident investigation?**
   Logs help investigators find out what happened and when it happened.
3. **Why might access to some logs be restricted?**
   Some logs may contain sensitive user or security information, so access is restricted to protect them.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

1. **Whether files/directories currently exist.**

   Yes. /tmp has several current files and folders used for temporary data, sockets, and sessionrelated information.

2. **Whether hidden files exist.**

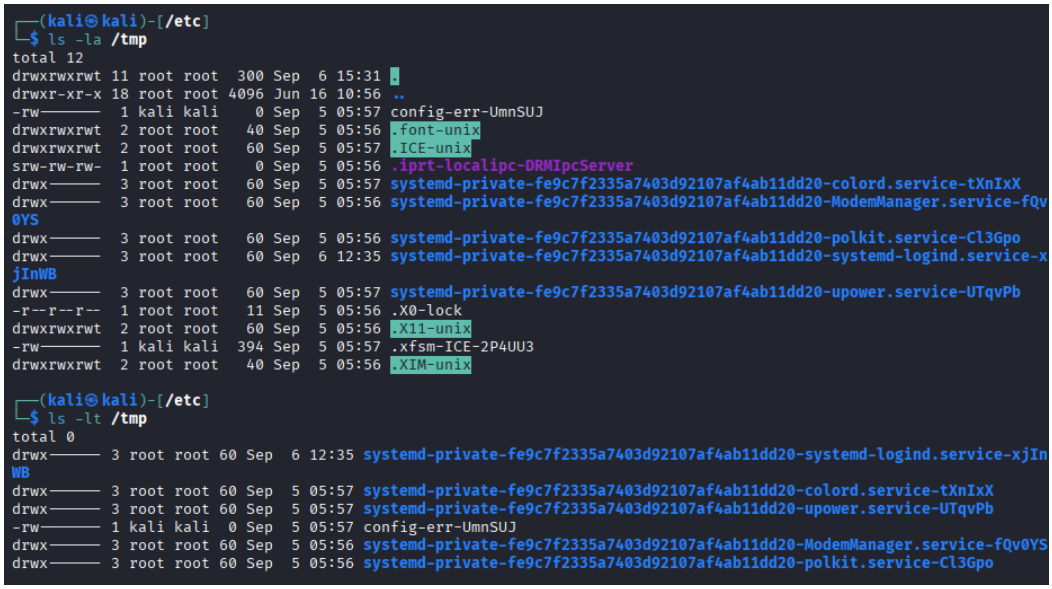
   Yes. It can include hidden items such as .X11-unix, .ICE-unix, and other dot-named entries.

3. **Who owns some of the entries.**

   Ownership varies by entry; some may belong to root and others to the user or process that created them.

4. **Which entries were modified recently.**

   ls -lt /tmp lists items by last modification time, with the newest entries at the top.



Answer:

Why can temporary directories be relevant during a cybersecurity investigation?

`/tmp` is important because many programs use it to store temporary files. In a security investigation, it may contain suspicious scripts, downloaded items, extracted archives, or malware-related files, so it should be checked closely.

Exercise 14 – Explore the `/proc` Filesystem

Investigate:

/proc

```

Examine:

cat /proc/cpuinfo cat

/proc/meminfo ``` 16

Then identify your current shell PID: echo

$$

```

Check whether a corresponding directory exists under `/proc`.

```
┌──(kali㉿kali)-[/etc]
└─$ cd /proc/

┌──(kali㉿kali)-[/proc]
└─$ cat cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 154
model name      : 12th Gen Intel(R) Core(TM) i7-12700H
stepping        : 3
microcode       : 0x432
cpu MHz         : 2688.004
cache size      : 24576 KB
physical id     : 0
siblings        : 2
core id         : 0
cpu cores       : 2
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2
 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3
fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgs
base bmi1 avx2 bmi2 invpcid rdseed adx clflushopt sha_ni arat md_clear flush_l1d arch_capabilities
bugs            : spectre_v1 spectre_v2 spec_store_bypass swapgs rfds bhi spectre_v2_user its
bogomips        : 5376.00
clflush size    : 64
cache_alignment : 64
address sizes   : 39 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id       : GenuineIntel
cpu family      : 6
model           : 154
model name      : 12th Gen Intel(R) Core(TM) i7-12700H
stepping        : 3
microcode       : 0x432
```

```
┌──(kali㉿kali)-[/proc]
└─$ cat meminfo
MemTotal:        2016604 kB
MemFree:          204772 kB
MemAvailable:    1164704 kB
Buffers:          118480 kB
Cached:           951680 kB
SwapCached:        28796 kB
Active:           655060 kB
Inactive:         896364 kB
Active(anon):     382952 kB
Inactive(anon):   121284 kB
Active(file):     272108 kB
Inactive(file):   775080 kB
Unevictable:         112 kB
Mlocked:             112 kB
SwapTotal:        976556 kB
SwapFree:         863564 kB
Zswap:                 0 kB
Zswapped:              0 kB
Dirty:                 4 kB
Writeback:             0 kB
AnonPages:        472004 kB
Mapped:           223624 kB
Shmem:             23184 kB
KReclaimable:      89140 kB
Slab:             138268 kB
SReclaimable:      89140 kB
SUnreclaim:        49128 kB
KernelStack:        7336 kB
PageTables:        14928 kB
SecPageTables:         0 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:     1984856 kB
Committed_AS:    3489096 kB
VmallocTotal:   34359738367 kB
VmallocUsed:       28504 kB
VmallocChunk:          0 kB
```

Answer:

1. **What information does `/proc/cpuinfo` provide?**

/proc/cpuinfo shows details about the CPU, including its model, clock speed, cache, and supported capabilities.

2. **What information does `/proc/meminfo` provide?**

/proc/meminfo gives information about the system's memory usage, including RAM and swap.

3. **Why do directories with numbers appear inside `/proc`?**

The numbered folders in /proc are process IDs for running programs.

4. **Is `/proc` an ordinary disk directory?**

No. /proc is a virtual filesystem created by the kernel, not a normal disk-based directory.

5. **Why could `/proc` be useful during security investigation?**

/proc is useful in investigations because it reveals active processes and system behavior, which can help spot suspicious activity.

Exercise 15 – Explore `/dev`

/dev

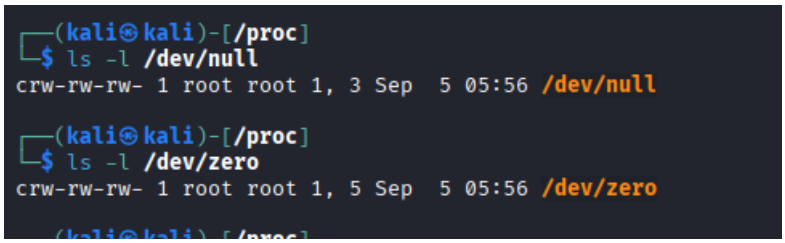
```

**Examine:**

**ls -l /dev/null ls –**

**l /dev/zero**

```



Answer:

1. **What does `/dev` represent?**

It is where Linux provides hardware and virtual devices as files. Programs can interact with devices such as disks and terminals through these files.

2. **Are the entries ordinary user files?**

No. They are character devices, not regular files. The c at the beginning of the permissions shows that they are character devices.

3. **Determine the file type of `/dev/null`.**

/dev/null is a character special device with major number 1 and minor number 3.

4. **Why does Linux expose devices through the filesystem?**

Linux treats devices much like files, which makes it easy for programs to interact with them. For example, /dev/null simply throws away anything sent to it, while /dev/zero keeps giving zeros whenever it is read.

Exercise 16 – Filesystem Space Investigation

Determine:

1. **Total filesystem capacity**

29G

2. **Used space.**

11G

3. **Available space.**

17G

4. **Percentage of filesystem utilization.**

40%

5. **Space occupied by your home directory.**

185M

6. **Space occupied by the `cyber_lab` directory.**

32K

Use appropriate combinations of:

df

df h

du

du -h

du -sh

Answer:

Explain the difference between `df` and `du`.

`df` shows disk space usage for an entire filesystem. `du` shows how much space a specific file or directory uses.

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported.

You are initially asked to examine only the filesystem.

Identify which of the following locations you would investigate and explain **why:**

/home

/etc

/var/log

/tmp

/proc
/dev

/root

Answers:

/home: useful for checking suspicious files, downloads, hidden items, scripts, and signs of user activity.

/etc: useful for spotting altered system settings, new accounts, sudo changes, or scheduled jobs.

/var/log: useful for reviewing logins, errors, and other traces of activity.

/tmp: may contain temporary malicious files or scripts used by an attacker.

/proc: shows currently running processes, which can reveal suspicious activity.

/root: if accessible, it may indicate that root access was obtained.